

GEOMIND-R

Generative inverse design of geopolymers for the immobilisation of radioactive Cs-137 and Sr-90

Second progress report — what has advanced since Checkpoint 1, the strengthened central chemical finding, the current data structure, and a prioritised list of the papers, datasets and data-types we now need you to help acquire.

Principal investigator	Mulham Fetna · ORCID 0009-0006-4432-798X
Supervisor	Dr. Abdulrazzaq Hammal · ORCID 0000-0003-1828-1376
Institution	[Institution / Department]
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Executive summary

What has changed, and the one thing that matters most

Since Checkpoint 1 an eleven-paper batch was mined into the database. The headline is a genuine strengthening of the project's central result: the evidence that *framework aluminium* governs radionuclide uptake is no longer a set of cross-sample correlations — it now includes a **within-sample causal test**.

141 Pool A rows (14 sources) — was 118	50 Pool B rows (7 sources) — was 16	33 audited findings — was 26	125 automated integrity tests passing
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THE RESULT TO TAKE AWAY

A companion study leached a series of geopolymers with water, stripping framework aluminium, and re-measured strontium uptake on the *same* samples. The samples that lost the most framework Al lost the most sorption: **$\text{corr}(\Delta[\text{Al}^{\text{IV}}], \Delta K_{\text{D}}) = +0.938$** . And the absolute density of framework-Al exchange sites, **$[\text{Al}^{\text{IV}}]$ (mmol g⁻¹), predicts K_{D} at +0.946** (+0.986 excluding one barely-reacted outlier) — against just **+0.19 for surface area**. This is the first evidence that tetrahedral Al^{IV} is the strontium exchange site, not merely a correlate of it.

THE THREE DEVELOPMENTS SINCE CHECKPOINT 1

- The framework-aluminium finding gained causal weight** (Section 2), and the descriptor is now ranked: absolute exchange-site density > per-silicon fraction.
- The immobilisation side independently reproduced the pattern**. Caesium leachability now stratifies cleanly by binder chemistry: metakaolin framework ≈ 13 > fly-ash framework ≈ 10 > calcium-rich slag ≈ 7.5.
- The obstacle is unchanged but sharper**. One specific paper — reporting NMR *and* a binding capacity together — remains the single highest-value acquisition, and Section 5 sets out exactly what to obtain (and acceptable substitutes).

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Section 1

What has advanced since Checkpoint 1

Eleven papers were triaged by content, then extracted inline and verified against source. Two were the specific sources flagged as needed at Checkpoint 1; both are now closed.

Paper	What it added	Effect
Varon <i>leached</i> (Open Ceramics 2026)	the within-sample causal test (Section 2)	Pool A +7 Sr K_{D}
Stanojević 2025	fly-ash GP; Cs passes the leach criterion yet >70% escapes	Pool B +2 Cs
Komljenović 2020	alkali-activated slag (a calcium system) — confirms the Ca precondition	Pool B +2 Cs
Qian 2001	adding aluminosilicate raises uptake even in a Ca binder	Pool A +4 Cs/Sr
Arbel-Haddad 2022	metakaolin framework; Cs cumulative-leached fractions	Pool B +6 Cs

Paper	What it added	Effect
Petlitchaia 2020 (foams)	bare framework out-performs a functionalised foam on raw capacity	Pool A +4 Cs
O'Donoghue 2026	structure only, no uptake target — <i>assessed, not ingested</i>	—
2 duplicates + 1 supplement	identified and kept out of the pools	—

Every value was read *in context* in the source. Where a source was imperfect it was flagged rather than fixed: a 2001 paper's unusual "mol g⁻¹" label on a distribution ratio was recorded as-is (magnitude indicates mL g⁻¹); an author-accepted manuscript with no DOI was given a flagged citation, not a guessed identifier; and one NMR-rich paper with no uptake measurement was deliberately left out rather than pad the dataset with rows no model can use.

Section 2

The strengthened central finding

Checkpoint 1 established that for cations, framework aluminium — the tetrahedral Al^{IV} that forms the charge-balancing exchange sites — predicts uptake, whereas surface area does not. The new batch turned this from correlation toward causation.

2.1 A within-sample causal test (finding F27)

A companion study to our Varon source re-measured the same seven geopolymers before and after 96 h of water leaching, tracking both framework aluminium and strontium uptake:

- [Al^{IV}] concentration predicts K_D:** corr = **+0.946** (n = 7), **+0.986** excluding one barely-reacted outlier — versus **+0.19** for BET surface area and **−0.84** for the Si/Al ratio.
- Leaching strips framework Al, and uptake falls with it:** corr(Δ[Al^{IV}], ΔK_D) = **+0.938**. The sample that lost the most framework Al (1.39 mmol g⁻¹) also lost the most K_D (2750 mL g⁻¹); the one that lost almost none lost the least.

TWO REFINEMENTS TO THE DESCRIPTOR

(i) **Absolute exchange-site density beats the fraction.** [Al^{IV}] in mmol g⁻¹ (+0.95) predicts better than Al^{IV} as a percentage (+0.80) or the per-silicon Al-richness index (+0.77) — capacity scales with *how many* exchange sites there are, not their proportion. (ii) The relationship is now **within-sample**, not only across samples — the strongest causal evidence a literature meta-analysis can provide short of new laboratory work.

2.2 The immobilisation side reproduces the pattern independently

The new leaching papers let caesium retention be compared across binder chemistries. It stratifies exactly as the framework-aluminium picture predicts:

Metakaolin framework	Cs leachability index ≈ 13.2–13.3 (Nevin)
Fly-ash framework	≈ 10 (Stanojević, Jang)
Calcium-rich slag (C-A-S-H)	≈ 7.0–7.8 (Komljenović)

The aluminosilicate framework holds caesium better than the calcium-silicate gel — the same conclusion as the adsorption side, reached through an independent property (leach resistance) and an independent set of papers. Qian's 2001 result closes the loop from the calcium side: adding aluminosilicate (metakaolin, zeolite) to a slag binder raises the Cs and Sr distribution ratio by 67–88%. More aluminosilicate exchange capacity, more uptake — in every direction tested.

2.3 The structural precondition still holds

The descriptor remains meaningful only where a condensed aluminosilicate framework exists. In calcium-rich binders the silicon network collapses to low-connectivity environments and the framework descriptor no longer applies — Komljenović's slag is exactly this regime, and its lower caesium retention is the visible consequence. The rule from Checkpoint 1 stands: never pool across structural classes.

Section 3

Current data structure

The two-pool architecture is unchanged; both pools grew. All data remains governed by a single machine-checked registry (17 assets, 33 findings, 16 decisions) with the three-axis trustworthiness labelling described at Checkpoint 1. Every row of both pools is listed in Appendices B and C.

POOL A — ADSORPTION (141 ROWS, 14 SOURCES)

Target: adsorption capacity or distribution coefficient. Of these, 75 are radionuclide (Cs/Sr) measurements.

Source	rows
oulu2026	63
literature_compilation	21
varon2025	7
varon_leached2026	7
lin2026	6
katada2024	6
tarnovsky2024	6
kurumisawa2021	5
xiang2021	5
petlitchaia2020	4
qian2001	4
baek2018	3
tian2019_espr	2
zhang2021	2

POOL B — IMMOBILISATION (50 ROWS, 7 SOURCES)

Target: leach resistance (leachability index, effective diffusivity, cumulative leached fraction). The batch added fly-ash, slag and metakaolin leaching sources and a new retention-type (cumulative leached fraction).

Source	rows
jang2016	14
nevin2026	10
kurumisawa2021	10
kim2026	6
arbelhaddad2022	6
komljenovic2020	2
stanojevic2025	2

Section 4

What we still cannot do, and why

The generative extension still cannot be trained, and the reason is the same structural one diagnosed at Checkpoint 1 — now with a sharper edge.

4.1 The descriptor is measured with a target only rarely

Framework aluminium predicts uptake, but solid-state NMR and a Cs/Sr capacity are seldom reported for the same samples. The new batch *improved* this — the causal test (F27) rests on seven samples measured twice — but it did not add new *compositions* carrying both the descriptor and a target. The paired evidence base is still a small number of sample families, and a generative model needs many more. This is a feature of how two research communities publish, not a gap our searching can close on its own — which is precisely why Section 5 asks for specific help.

4.2 The faithful replication is still blocked on confidential data

The reference model was trained on 112 samples from one laboratory; only ten are public. A faithful replication needs the full set, which is confidential and must be requested.

4.3 Inverse design is still out of reach on present data

There are 75 radionuclide rows in the adsorption pool, spread across many laboratories and protocols. Training a generative inverse-design model on so few rows from so many sources would learn inter-laboratory differences and report them as chemistry. The defensible near-term output remains a descriptor-and-methodology study — now materially stronger thanks to the causal result — not a generative model.

Section 5

Required acquisitions & data needs (the requests)

This section is the practical ask. It lists, in priority order, the specific papers and dataset to obtain, and — importantly — **what an acceptable substitute looks like**, so that if an exact source is inaccessible, an equivalent one can be found that yields the same insight. Please help **download** these where institutional access allows, or identify **replicas** / **drop-in replacements** matching the stated criteria.

5.1 Specific papers to obtain

Priority	Source	Identifier	Why it matters	Unblocks
1 — critical	Geddes <i>et al.</i> 2025 — <i>Alkali-mediated Sr incorporation mechanism and binding capacity of alkali aluminosilicate hydrate in geopolymers</i>	J. Hazard. Mater. 488 (2025) 137426 · DOI 10.1016/j.jhazmat.2025.137426 (as cited in Kim 2026)	The rare paper that reports NMR mechanism and a binding capacity together, on the same samples — exactly the descriptor-plus-target bridge the whole project turns on. Same Sheffield group whose work already underpins our strongest findings.	M4, M5
2 — high	<i>Next Sustainability</i> 2025 (Cs/Sr sorption on activator-concentration series)	Article S2950631X25000024 (paywalled; DOI not yet resolved)	The only source found with a 1–5 M activator-concentration axis carrying Cs <i>and</i> Sr <i>and</i> NMR together — a second composition-varying bridge.	M4, M6

5.2 The confidential dataset (request, not download)

Priority	Dataset	How to obtain	Why it matters	Unblocks
1 — critical	GEOMIND 112-sample training dataset	Written request to the authors of the reference paper (RSC <i>Digital Discovery</i> , DOI 10.1039/d5dd00383k). A drafted letter already exists in the project.	The only route to a <i>faithful</i> replication of the reference model — 112 samples from one laboratory under one protocol, larger than every literature avenue combined.	M3a

5.3 Data-types needed — and what counts as an acceptable substitute

Beyond the two named papers, the project needs specific *kinds* of data. For each, the right-hand column states the minimum an alternative source must provide to give the same insight — use these as search criteria when a named paper cannot be obtained.

Data-type we need	Minimum acceptable form (a valid drop-in substitute)
Descriptor + target, same samples — the core need	Any study giving a ²⁹ Si Q ⁴ (mAl) deconvolution <i>or</i> a ²⁷ Al Al ^{IV} fraction and a Cs or Sr capacity / K _d / leachability index, measured on the same ≥4 samples that differ in Si/Al . Material family is flexible (metakaolin, fly-ash, or slag) provided it is stated.
A leaching / degradation series	Any dataset measuring framework Al and uptake on the same samples <i>before and after</i> some treatment (leaching, ageing, thermal) — this is what makes the causal test (F27) possible; a second independent instance would greatly strengthen it.

Data-type we need	Minimum acceptable form (a valid drop-in substitute)
A single-lab, many-sample formulation set	≥50 geopolymer formulations from one laboratory/protocol with measured properties (density, strength, viscosity, and ideally a Cs/Sr metric) — the minimum scale at which a generative model can be trained without learning inter-lab artefacts.
27Al MAS NMR with sorption	Direct Al ^{IV} /Al ^V /Al ^{VI} quantification paired with a Cs/Sr metric — a cleaner descriptor than inferring Al ^{IV} from ²⁹ Si, if available.

5.4 Where such data is most likely to be found

Two research groups publish almost all of the paired descriptor-and-target data this project depends on, and their other papers are the best hunting ground:

- The **Sheffield group** (Walkley, Provis, Stennett, Geddes, Nevin, O'Donoghue, Stanojević, Komljenović) — solid-state NMR of Cs/Sr geopolymer wasteforms.
- The **CEA Montpellier group** (Poulesquen, Varon, Gossard, Barré, Petlitckaia) — strontium sorption on geopolymers with NMR and K_D.

Useful venues: J. Hazard. Mater. , J. Nucl. Mater. , Cement & Concrete Research , Dalton Transactions , Open Ceramics , and Construction & Building Materials . Open datasets on Zenodo / figshare tagged "geopolymer NMR" or "caesium sorption" are worth a scan for machine-readable tables.

Section 6 Plan and requests

6.1 What proceeds independently

The descriptor-and-methodology result is now strong enough to write up: framework aluminium predicts cation uptake (with a within-sample causal test), absolute exchange- site density is the right form of the descriptor, the effect has a stated structural precondition, and a simple saturation screen exposes published capacities that are extrapolation artefacts. None of this needs the confidential dataset.

6.2 What we ask of this checkpoint

- Acquire the Section 5.1 papers** — above all Geddes 2025 (institutional access to J. Hazard. Mater.), or identify an equivalent meeting the Section 5.3 criteria.
- Support the confidential-data request** to the reference-model authors, or advise on the right channel.
- Guidance on a small own-laboratory NMR measurement** on a few sorbents of known capacity — the one route that would generate genuinely new paired data rather than re-using the literature, and would let us replicate the causal test in-house.
- Agreement** that the descriptor-and-methodology paper is the right near-term scientific target.

Appendix A Full finding register

All 33 audited findings, most-recent last, each with the evidence that established it and the action taken. Severity is the data-integrity impact.

critical F1 — Si_Al and Density transplanted from the GEOMIND paper [resolved] EVIDENCE 9 of 11 unique Si_Al values match the paper's 10 samples exactly to 3 dp; all 10 published density values appear; 91 of 92 mixes carry a single broadcast value; neither column exists in either upstream source file. ACTION Both columns dropped in clean.py. Must never be reinstated as features.
critical F2 — Dataset far smaller than 867 rows implies [resolved] EVIDENCE Only 92 distinct mix designs; Metakaolin constant zero; features have 2-6 unique values. ACTION Grouped CV by mix design is mandatory. Metakaolin dropped.
high F3 — Adsorption data has extensive internal duplication [resolved] EVIDENCE 24 Qmax_Cs rows contain only 16 distinct values; Qmax_Sr has 7 rows. ACTION Repeats flagged (not deleted) via cs/sr_value_repeated.
medium F4 — Redundancy and version disagreement [partially resolved] EVIDENCE CSV/XLSX byte-identical; 18 duplicate rows; v2 disagrees in 22 cells. ACTION CSV declared canonical; duplicates removed; v2 unresolved.
low F5 — Metadata.csv is cp1256, not UTF-8 [resolved] ACTION Handled in the ingestion layer.
critical F6 — Feasibility_Ranges is mislabelled and anti-correlated with truth [resolved with open gap] EVIDENCE Rejects 7 of 9 feasible published samples, accepts the 1 infeasible. Its constants are verbatim from the retired tool at tag v1.0.0. ACTION Sheet not read. Provisional envelope used instead, flagged provisional=True.

critical F7 — Compiled data failed source audit - 10 of 31 rows **[resolved]**

EVIDENCE

(a) 3 rows labelled Sr are Eu-152+154. (b) 4 Niu 2022 rows claim Langmuir Qmax from a paper tabulating none. (c) Na-MK misattributed from another paper. (d) K-MK is a verbatim copy of K-MKBFS - a sorbent that does not exist.

ACTION

Quarantined/relabelled/re-typed as named constants in code.

medium F8 — Unmined master sheets add nothing **[resolved]**

EVIDENCE

Adsorption_Isotherms and Cs_Isotherms duplicate Functional_Data with the same fabricated rows; only 3 new values, which are cobalt.

ACTION

Marked duplicate. Not ingested.

high F9 — Surface area does not predict CATION uptake **[open]**

EVIDENCE

Cation-only, per source - lin2026 -0.90 (n=4), xiang2021 -0.90 (n=5), oulu2026 -0.47 (n=21), tarnovsky2024 +0.22 (n=6, inconclusive). Pooled cations +0.08. Varon: BET +0.19 vs Al-IV +0.95.

ACTION

Al-IV is the candidate primary feature. NOTE - an earlier claim of "six datasets confirm a negative BET correlation" was an OVERSTATEMENT, corrected here.

high F10 — Pooling adsorbates flips the sign of the BET relationship **[resolved]**

EVIDENCE

cations corr(BET,capacity) = +0.08; dyes = +0.66.

ACTION

Never pool adsorbate classes. Dyes use surface area; cations use charge sites. pooling_warning() enforces the check.

high F11 — Sr_Immobilization_Mechanisms is untraceable and largely duplicated **[resolved]**

EVIDENCE

Audit 2026-07-20 - the ONLY master sheet with no Source column at all, plus 66.7% internal duplication among its numeric cells. Same signature as the fabricated Na-MK/K-MK rows in F7.

ACTION

Relabelled FALSE / DISCARD in the warehouse. Not usable.

medium F12 — The v2 "conflict" was precision, not disagreement **[resolved]**

EVIDENCE

All 22 differing cells are Comp_Strength in rows 843-866 and differ only by rounding (34.36 vs 34.35777). N data.xls genuinely reports up to 6 dp, and 18 of the 24 v2 values appear verbatim upstream. So v2 preserved source precision and Physical_Data rounded to 2 dp.

ACTION

Q1 closed. v2 status suspect -> verified. Physical_Data remains canonical; the difference (<=0.005 MPa) is immaterial for modelling.

medium F13 — Several context sheets cite papers we do not hold **[open]**

EVIDENCE

Audit 2026-07-20 - Kim_2026, Yildirim_2024 and Duque_Redondo_2023 are cited by Surface_Complexation_Sr, MD_Simulation_Sr-Cs, Zeolite_Comparison_Sr-Cs, Sr_Leaching_Data, EXAFS and XRD sheets. None are in papers/references. Surface_Complexation cites "Eq. (10)-(12)" - equations, not sources, so those rows are model output rather than measurement. FTIR is 88.9% duplicated, XRD 72.7%.

ACTION

All remain UNVERIFIED and excluded from modelling. Acquiring the three papers is the only way to promote them; recorded as Q6.

high F14 — A high R2 Langmuir fit can leave Qmax essentially unconstrained **[open]**

EVIDENCE

Baek 2018 chabazite: Q0L = 1249.5 mg/g with K_L = 7.9e-05 L/mg and R2 = 0.99989, while the paper's own kinetics report equilibrium uptake of just 1.184 mg/g. K_L is two orders of magnitude below the other two zeolites in the same table. At low coverage the Langmuir isotherm is near-linear, so it fits almost perfectly while the plateau is extrapolated ~1000x beyond any measured point.

ACTION

Ingested but flagged - it exceeds the schema bound of 1000 mg/g and validate() reports it. Must be excluded from any fit unless the extrapolation is defended. Generalises: R2 is not evidence that a fitted Qmax is meaningful. Screen future Langmuir rows on K_L and on the measured concentration range, not R2 alone.

critical F15 — Fitted capacities cannot be validated - C0 was never captured **[resolved with open gap]**

EVIDENCE

Applying the F14 screen to all 35 fitted-isotherm rows: 32 of the 33 rows with a usable K_L have NO recorded initial concentration, so Langmuir saturation ($\theta = K \cdot C_0 / (1 + K \cdot C_0)$) cannot be computed. The screen runs on exactly one row (GF600, $\theta = 0.93$). Chabazite was caught only because its paper stated the equilibrium uptake in prose.

ACTION

initial_conc_mg_L made required for fitted-isotherm rows going forward; existing rows flagged c0_unknown. PROGRESS 2026-07-20: C0 recovered for Baek (2658 mg/L) and El-Naggar (1000 mg/L), taking screenable rows from 1 to 10 - of which 8 are sound, 1 borderline (stilbite, θ 0.775) and 1 a confirmed artifact (chabazite, θ 0.174). 25 rows still lack C0. COMPLETE 2026-07-20: C0 recovered for 7 sources; 29 of 35 rows now screenable. 27 sound, 1 borderline (stilbite 0.775), 1 artifact (chabazite 0.174). Six rows remain unscreenable because their methods state no isotherm range. Lesson recorded THREE times: K_L alone is not a screen - saturation is $K \cdot C_0 / (1 + K \cdot C_0)$, so a small K is compensated by a large C0. The same instinct would have discarded 19 sound rows.

medium **F16 — theta = 1 - R_L, so the saturation screen needs no C0** [resolved]

EVIDENCE

The Langmuir separation factor $R_L = 1/(1+b \cdot C_0)$ is reported by many papers as a favourability check. Our saturation measure $\theta = b \cdot C_0 / (1 + b \cdot C_0)$ is exactly $1 - R_L$. Verified against the independent C_0 route on six El-Naggar rows: the two derivations agree to three decimals. Resolved Tian ESPR (θ 0.944 Cs, 0.994 Sr) with back-calculated C_0 5.25 and 9.77 mmol/L, matching the Fig. 5 axes.

ACTION

Use R_L as the PRIMARY screen and C_0 recovery only as fallback. Found by rendering PDF figures and reading them visually - a capability previously and wrongly described as "manual figure reading" and deferred to Mulham.

high **F17 — Zhang 2021 rows are extrapolations with an unphysical intercept** [resolved]

EVIDENCE

Figure 8 read visually. The slope reproduces the published $Q_{\max}$ exactly ($1/11.196 = 89.32$; $1/22.464 = 44.52$), but both intercepts are NEGATIVE (-0.5365, -2.0942), which in $C_e/q_e = 1/(Q_{\max} \cdot b) + C_e/Q_{\max}$ implies $b < 0$ and is unphysical; the table nonetheless reports positive b . Data reach only $C_e = 3.45$ and 3.75 mg/L, giving $\theta = 0.365$ and 0.074 - the slag fit reaches 7% of its plateau, worse than the chabazite artefact at 17%.

ACTION

Both rows flagged as extrapolations and excluded from modelling alongside chabazite. Screen now COMPLETE at 35 of 35: 31 sound, 1 borderline, 3 extrapolated.

high **F18 — The Oulu 29Si deconvolution was already published; it explains our failed fit** [resolved]

EVIDENCE

Table 2 of Hossain 2026 (doi:10.1016/j.matdes.2026.115471, page 6) reports the deconvolved Si environments for all 12 selected samples - the values our own fit attempted to derive. Found by rendering the page and reading it. The authors use TEN environments (Q0, Q1, Q2(1Al), Q2(0Al), Q4(4Al), Q3(1Al), Q4(3Al), Q3/Q4(2Al), Q3/Q4(1Al), Q4(0Al)) including Q3 and Q2 species with non-bridging oxygens. Our model used FIVE Q4(mAl) Gaussians only, which cannot represent that intensity; the optimiser pushed it outward and drove Q4(3Al) to zero.

ACTION

Confirms the NMR rejection was correct AND for the right reason - the chemical sanity check detected a structurally wrong peak model that $R^2 = 0.998$ and both pre-registered gates could not. Pipeline closed. The Oulu letter is no longer needed for this purpose. Values transcribed into src/geomind/data/nmr_ari.py; 11 of 12 rows sum to exactly 100%, one to 99.0 (published as such).

high **F19 — The Al-richness descriptor is valid only within a structural class, and the Oulu backfill does not widen its evidence base** [resolved]

EVIDENCE

Backfilling ARI from F18 attached the descriptor to 39 pool rows / 12 sorbents. Split by structure, the 12 sorbents behave as two different materials. Ca-FREE ($n=4$, Q4 45-73%): ARI +0.932, framework-only ARI +0.940 - uptake spans 2.13-11.53 mg/g (sd 3.98). Ca-BEARING ($n=8$, Q4 0-26%, Q0-Q2 dominated): ARI -0.108, Si/Al -0.135, Ca/Al -0.104, Q4% -0.247, BET +0.462 - NOTHING predicts, and uptake is nearly flat at 7.27 +/- 1.31 mg/g regardless of composition. Pooling all 12 gives ARI +0.536, a mediocre value that conceals two regimes. Ca acts as a network modifier: there is no condensed aluminosilicate framework for a framework descriptor to measure, and uptake proceeds by a different mechanism (C-A-S-H exchange), roughly composition-independent.

ACTION

Restrict the ARI claim to Q4-rich framework aluminosilicates and state the structural gate explicitly. Generalises F9 one level down: never pool ADSORBATE classes, and never pool SORBENT structural classes either. IMPORTANT correction to my own estimate - I projected this backfill would widen the descriptor to ~19 sorbents; in fact only 4 of the 12 lie in the regime where it applies, so the evidence base is unchanged at Varon $n=7$ (Sr) + Oulu $n=4$ (NH4+). Al^{IV} scarcity remains the binding constraint on M4/M5.

critical **F20 — NMR characterisation and adsorption-capacity measurement are near-disjoint in this literature - Al^{IV} scarcity is structural, not a sampling accident** [open]

EVIDENCE

Corpus-wide classification of all 30 PDFs by keyword balance (adsorption terms vs leaching/immobilisation terms) against NMR content. Of the 6 papers carrying substantial solid-state NMR, FOUR are immobilisation/wasteform studies (Nevin 2026 nmr=54, Kim 2026 nmr=66, Vettese 2025 nmr=33, Kurumisawa 2021 nmr=13), ONE is adsorption (Hossain/Oulu nmr=11, but NH4+ and dyes only - no Cs/Sr), and ONE is mixed (Varon 2025 nmr=32). Meanwhile ALL 13 papers classified as adsorption studies report ZERO solid-state NMR. Varon is the single bridge between the two - which is exactly why Al^{IV} occurs in precisely 7 pool rows, all from Varon. These are two different research communities: water-treatment groups measure isotherms and BET and never run MAS NMR; nuclear-wasteform groups run full 29Si/27Al/133Cs NMR and measure leach rates and diffusivities, never isotherms.

ACTION

More scraping CANNOT fix Al^{IV} scarcity - the co-occurrence is rare by the structure of the field, not by chance, so Q7 has no answer within the adsorption literature. Two routes remain: (a) measure NMR ourselves on sorbents of known capacity, or (b) REFRAME the M4/M5 target property from adsorption capacity to leach resistance, where NMR is abundant. Note that route (b) reverses earlier triage - Kurumisawa 2021 and J. Environ. Manage. 370 (2024) were both REJECTED as "diffusivity/leaching studies", i.e. we discarded precisely the papers richest in the descriptor we need. The project aim ("immobilisation of Cs-137/Sr-90") is arguably better served by leach resistance than by adsorption capacity anyway; the two have been conflated. Decision required from PI - see reports/q7-descriptor-hunt-report.md.

high **F21 — Sr_Leaching_Data audited against source for the first time - LI verified, De corrupted to zero, and the Yildirim rows are misattributed** [resolved]

EVIDENCE

Both cited sources (Kim 2026, Nevin 2026) are now known to be on disk (see F13), so this sheet became auditable. Line-by-line against Kim Table 2 (p10) and Nevin p6: (a) LEACHABILITY INDEX - VERIFIED, 6 of 6 determinable rows exact. Kim NaSr1 15.5, NaSr2 14.8, NaSr3 16.0, K Sr3 16.0; Nevin Sr_1 18.7, Sr_3 19.1. K Sr1/K Sr2 are correctly recorded as below detection - Kim states Sr leachate was under the ICP-OES limit (0.001 ppm), yielding non-detectable De and LI. (b) De_cm2_s - CORRUPTED. Every row reads 0.0, whereas Kim Table 2 reports 8.88E-16, 3.80E-14, 2.10E-16, 6.36E-16, 5.80E-18 and so on. Precision was lost to zero. Kim's De is also TIME-RESOLVED over five points (2 h, 7 h, 7 d, 14 d, 21 d), so no single scalar De pairs with the averaged LI in the first place. (c) CN_Sr_EXAFS on the two Yildirim_2024 rows - MISATTRIBUTED. The sheet gives Sr_Pre_1 = 7.87 and Sr_Post_1 = 7.95 sourced to Yildirim_2024. Kim p26 states the NaSr3 coordination number "was 7.87 +/- 0.69 before the leaching test" and "7.95 +/- 0.43 afterwards" - these are Kim's own NaSr3 pre/post values, and Kim attributes the pre-leach figure to reference [7], which is Geddes et al., J. Hazard. Mater. 488 (2025) 137426 - NOT Yildirim. The K Sr3 value 7.55 is correctly attributed.

ACTION

Promote the LI column to VERIFIED_TRUE - it is the first immobilisation target variable in the project to survive source audit. Quarantine De_cm2_s entirely (re-extract from Kim Table 2 as a time series if needed). Quarantine both Yildirim_2024 rows as misattributed - third instance of the F7 pattern.

high **F22 — Pool B built and audited - LI verified twice over, but it has almost no compositional variation paired with NMR** **[resolved]**

EVIDENCE

Pool B v1 = 16 rows / 2 sources (Kim 2026, Nevin 2026), of which 8 carry a retention target, 2 are left-censored, and 6 are structure-only. (a) The LI = -log10(De) relation is now confirmed against TWO independent published tables - Kim Table 2 (12 of 13 cells to within 0.06) and Nevin Table 2 (4 of 4: 18.66/19.11/13.27/13.22 vs published 18.7/19.1/13.3/13.2). (b) A PUBLISHED ERRATUM was found. Kim Table 2 cell NaSr1 @ 2 h prints "8.88 E-16 (15.5)" but -log10(8.88e-16) = 15.05. Kim's own column Average LI of 15.5 equals mean(15.05, 15.94) exactly; had the cell truly been 15.5 the average would read 15.7. The printed (15.5) is a typo for (15.1). Verified at 430 dpi, so it is not a rendering artefact. The column average is unaffected. (c) The master sheet Sr_Leaching_Data carried only Nevin's two Sr rows and SILENTLY DROPPED both Cs rows (Cs_1 LI 13.3, Cs_3 LI 13.2) - a 50% loss of that table, undetectable without reading the source. (d) Nevin's headline result is preserved: in the SAME K-A-S-H gel, Sr is retained roughly six orders of magnitude better than Cs (De 7.8e-20 vs 6.0e-14). Nuclide identity dominates structure entirely at fixed composition.

ACTION

LIMITATION, stated plainly: only 4 rows pair an NMR descriptor with a retention target, and across those 4 the ARI is nearly constant (2.738-2.855) because Nevin is ONE formulation at Si/Al = 3. Kim varies activator (Na vs K) and loading but reports no Q4 deconvolution. So Pool B is currently as descriptor-starved as Pool A, for the opposite reason - Pool A has composition without NMR, Pool B has NMR without composition. Neither supports a descriptor model yet. This raises the priority of Q8 (Geddes 2025) further: it is the only identified source that may carry composition, NMR and a retention/binding measure together.

medium **F23 — Not every published deconvolution assigns m uniquely - Vettese's ARI is indeterminate and must not be computed** **[resolved]**

EVIDENCE

Vettese 2025 Table 4 (p7) reports a 29Si deconvolution, but its first two columns carry AMBIGUOUS assignments: "Q3(1Al) or Q4(4Al)" (m = 1 or 4) and "Q3 or Q4(3Al)" (m = 0 or 3). Computing the Al-richness index therefore has two permitted answers for the unleached sample: ARI = 2.289 if the Q4 readings are taken, or 0.861 if the Q3 readings are - a factor of 2.66. Populations do sum correctly (100.0 / 100.1 / 100.1), so the table is internally sound; it is the ASSIGNMENT that is undetermined, not the fit. Hossain and Nevin both assign m uniquely; Vettese does not.

ACTION

Do NOT compute or store an ARI for Vettese. A pipeline that derived ARI from any Q^n(mAl) table without checking assignment uniqueness would silently inject a 2.7x error into the one descriptor the project depends on. Vettese can still contribute structure-only rows (raw populations) and its Cs leach data, which is figure-only and single-formulation. Any future adapter must carry an assignment-ambiguity flag.

critical **F24 — Kurumisawa 2021 recovered - a rejected paper that bridges both pools - and it exposes a 12-order-of-magnitude naming collision in "diffusion coefficient"** **[resolved]**

EVIDENCE

Kurumisawa 2021 (doi:10.1617/s11527-021-01758-y) was REJECTED during batch triage as "a diffusivity study (diffus 65)". Reading it inline shows that verdict was wrong on content: (a) Figure 3b publishes the Langmuir MONOLAYER Cs CAPACITY for five metakaolin geopolymers made with five different alkali activators (K11 0.315, K9 0.425, 0.66K 0.530, N11 0.727, KN11 0.393 mmol/g) - a Pool A capacity WITH real compositional variation, which Pool A is short of. Pool A: 121 -> 126 rows. (b) Figure 2 publishes a Cs diffusivity for the same five specimens - Pool B. Pool B: 16 -> 26 rows. (c) THE COLLISION. Kurumisawa's is an INGRESS coefficient (Cs diffusing INTO an immersed specimen), reported in m2/s and of order 4e-12 m2/s = 4e-8 cm2/s. Kim's and Nevin's are ANSI/ANS 16.1 EFFECTIVE diffusivities (nuclide leaching OUT of a doped wasteform), of order 1e-16 and 1e-19 cm2/s. Same name, opposite experiment, TWELVE orders of magnitude apart. (d) Table 3 is the 29Si NMR of the ACTIVATOR SOLUTION (Q0-Q3 of the silicate), NOT of the hardened gel - so this paper supplies no framework Al^IV despite being NMR-bearing. (e) Applying the F14/F16 saturation screen to the recovered capacities: K11 theta 0.962 sound, KN11 0.847 sound, K9 0.704 borderline, 0.66K 0.608 borderline, and N11 theta 0.499 an EXTRAPOLATION ARTEFACT - N11 being the HIGHEST reported capacity of the five.

ACTION

Both halves ingested through separate adapters under D13. RetentionType gains INGRESS_DIFFUSIVITY as a distinct member and pooling_warning() now raises an explicit CATEGORY ERROR if both diffusivity kinds appear together. N11 flagged as an artefact. All Kurumisawa rows carry from_figure=True. LESSON: the batch triage rejected this paper on keyword balance without reading it, and keyword balance cannot see a capacity that lives only in a figure. The other triage rejections should be re-read the same way - starting with J. Environ. Manage. 370 (2024).

medium **F25 — The last triage rejection (J. Environ. Manage. 370) is upheld as data, but it is a high-value BIBLIOGRAPHY that names the paired-data sources both pools lack** **[resolved]**

EVIDENCE

D15 re-read of Mukiza et al., J. Environ. Manage. 370 (2024) 122746, "Recent advances in immobilization of Cs/Sr-bearing wastes in AAMs: a review". It is a REVIEW, not primary data: zero Langmuir/Qmax/Kd of its own, and its Table 2 explicitly states "the data are not directly comparable". So the batch rejection was CORRECT for direct ingestion - none of its tabulated numbers may enter either pool as primary rows. BUT reading it surfaced exactly the paired-descriptor sources F20/F22 said were rare: (a) Komljenovic et al. 2020, J. Hazard. Mater. 388, doi:10.1016/j.jhazmat.2019.121765, "Immobilization of cesium with alkali-activated blast furnace slag" - the review states it ties 29Si NMR speciation (Q1, Q2(0Al), Q2(1Al) sites, Al^IV/Si of C-A-S-H) to strength AND Cs. NMR + composition + Cs in one paper. Same group (Provis) as Nevin/Geddes. A second candidate F20 bridge alongside Geddes 2025. (b) Qian et al. 2001, J. Nucl. Mater. 299, 199-204, doi:10.1016/S0022-3115(01)00700-0 - Cs and Sr distribution ratios R_d (3.9e3->6.4e3 mol/g Cs; 6.4e3->1.2e4 mol/g Sr). A K_d measure = Pool A. (c) Jang et al. 2016, J. Hazard. Mater. 318, 339-346, doi:10.1016/j.jhazmat.2016.07.003, "Physical barrier effect ... on diffusivity of Cs and Sr" - a Cs/Sr diffusivity (Pool B) source, and it is ALREADY ON DISK (papers/references/Physical barrier effect ...pdf), registered but never mined.

ACTION

Keep the review out of both pools as a data source. Use it as a source index. Q9 opened for Komljenovic 2020 (acquire), Q10 for Qian 2001 (acquire), and Jang 2016 queued for Pool B extraction since it is already held. This is the third time (after F24 Kurumisawa and the Kim/Nevin content-index miss) that real value was found by reading a document already in hand rather than acquiring a new one.

medium **F26 — Jang 2016 mined into Pool B - 14 rows, both nuclides, with pore structure; and it confirms the LI aggregation trap a third time** **[resolved]**

EVIDENCE

Jang, Park & Lee, J. Hazard. Mater. 318 (2016) 339-346, doi:10.1016/j.jhazmat.2016.07.003 - the Q11 source, already on disk, now extracted. Seven doped waste forms (F1/F2/F3 fly-ash geopolymers, S1/S3/S5 fly-ash+slag blends, PC Portland baseline), each with Cs AND Sr leachability indices (Table 7) and pore structure (Table 5). Pool B: 26 -> 40 rows / 4 sources. (a) LI AGGREGATION CONFIRMED a third time. Jang's LI = mean over intervals of -log10(De_n): F1 Cs interval De's give 10.02 ~ published 10.0, whereas -log10(mean De) = 9.46 would be wrong by 0.5. Vindicates schema trap 4. (b) Sr LI > Cs LI in EVERY one of the 7 specimens - the universal Cs<Sr retention result, now across 7 more matrices, consistent with Nevin's ~6-order gap. (c) Si/Al deliberately NOT stored. The precursor-bulk Si/Al is computable (1.55-2.13) but omits the sodium-silicate activator Si, so it is not the gel Si/Al, and the F/S series use different silicate loadings (0.25 vs 0.17) so it is not even comparable across specimens. Storing it would repeat the F1 transplant error in a new form. Recorded as provenance-only constants. (d) Three new pore-structure columns added (porosity, capillary pore volume, critical pore diameter); Jang links these to De. corr(porosity, Cs LI) = -0.27 across the 6 geopolymers - weak, consistent with Jang's own point that critical pore DIAMETER matters more than total porosity.

ACTION

Ingested via _from_jang2016 as 14 LI rows. Mean De values retained as documented adapter constants (uncorrupted, unlike Kim's) but not emitted as separate rows, to avoid implying an independent measurement - a diffusivity target can regenerate them later. PC baseline flagged matrix_class=portland_cement, trivially filterable. Q11 closed. Fourth on-disk recovery of the session.

critical **F27 — Varon's leaching companion paper provides a WITHIN-SAMPLE causal test - framework Al^{IV} concentration governs Sr K_D, and destroying it by leaching destroys sorption** **[resolved]**

EVIDENCE

Batch-4 paper Varon et al., Open Ceramics 25 (2026) 100895, doi:10.1016/j.oceram.2025.100895 - the leaching companion to varon2025 (Cleaner Materials). Its 0 h data are IDENTICAL to varon2025 (verified: K-0.5-16 BET 72, AlIV 96.0%, KD 4100) and are NOT re-ingested; the 96 h water-leached state is new. Tables 3-4 give, for all 7 samples at 0 h and 96 h: BET, [AlIV]_exp (mmol/g), AlIV/V/V_I %, full Q4(mAl) speciation, Sr KD and Qmax. RESULT: [AlIV] CONCENTRATION (mmol/g) is the strongest predictor of Sr KD: corr = +0.946 (n=7), rising to +0.986 excluding the barely-reacted outlier K-1.2-8 (H₂O/M₂O=8, BET 15, Qmax 0.02). This beats AlIV% (+0.80), ARI (+0.77) and crushes BET (+0.19, or -0.24 without the outlier) and Si/Al (-0.84). THE CAUSAL TEST: leaching removes framework Al, and corr(Δ[AlIV], ΔKD) = +0.938 - the samples that lose the most framework Al on leaching lose the most sorption. K-0.5-16 loses 1.39 mmol/g AlIV and 2750 mL/g KD (largest of both); K-1.5-16 loses 0.04 mmol/g AlIV and the least KD. This is the same sample losing capacity AS it loses exchange sites - close to a causal demonstration that tetrahedral Al^{IV} is the Sr exchange site, not merely a correlate.

ACTION

Ingested 7 NEW 96h-leached Sr KD rows via _from_varon_leached2026 (Pool A 126 -> 133 rows / 12 sources), sorbent_name "... (leached 96h)" so they can never be pooled with the fresh varon2025 rows. The "Interplay..." PDF in the same batch is a confirmed DUPLICATE of varon2025 (skipped). Strengthens the project's central finding: framework Al predicts cation uptake, and this is the first WITHIN-SAMPLE (not just cross-sample) evidence for it. Note [AlIV] CONCENTRATION > AlIV% > ARI as the ranking - absolute exchange-site density beats the per-Si fraction.

medium **F28 — Stanojevic 2025 - 2 Cs leachability rows for a fly-ash geopolymer; Cs passes the LI criterion yet >70% leaches, reaffirming Cs mobility** **[resolved]**

EVIDENCE

Batch-4. Stanojevic et al., Constr. Build. Mater. 500 (2025) 144217, doi:10.1016/j.conbuildmat.2025.144217 (Komljenovic/Provis group). Fly-ash geopolymer, Cs doped at 2% and 5%, semi-dynamic leach. Lmean (Fig. 6b) = 10.3 (2%Cs) and 9.9 (5%Cs) over 90 days (9.7/9.3 over the first 5 days) - both clear the ANSI/ANS-16.1 / NRC floor of 6. BUT cumulative Cs leached exceeds 70% after 5 days (71.0% / 76.5%): the LI criterion is passed while most of the Cs still leaves - Cs remains highly mobile. 29Si shows Q4(mAl) (fly-ash framework), and the higher 5% Cs loading introduces less-crosslinked Q1,2 units (Cs slows condensation), a qualitative echo of the framework story.

ACTION

Ingested 2 Pool B rows via _from_stanojevic2025 (40 -> 42 rows / 5 sources), from_figure=True (Lmean read from the Fig. 6b annotation). One fly-ash geopolymer at two Cs LOADINGS (a control variable), so no composition variation; the Q4 data is figure-only and not digitised into per-sample descriptor rows. Reinforces the universal Cs<Sr pattern: even a passing LI hides >70% Cs release.

medium **F29 — Komljenovic 2020 (Q9) closed - it is a Ca-system (C-A-S-H) and QUANTITATIVELY confirms F19 - the aluminosilicate framework immobilises Cs better than a Ca gel** **[resolved]**

EVIDENCE

Batch-4, Q9 acquisition. Komljenovic et al., J. Hazard. Mater. 388 (2020) 121765, doi:10.1016/j.jhazmat.2019.121765. Alkali-activated blast-furnace slag (AABFS), Cs doped 2%/5%. The hoped-for "framework NMR + Cs" bridge it is NOT: AABFS is a C-A-S-H binder whose 29Si NMR shows only low-connectivity Qn(mAl) sites (n=1-3, m=0-1) with NO framework Q4(mAl) - exactly the Ca-modifier regime of F19, so the descriptor is C-A-S-H chain length / AlIV/Si, not ARI. Cs preferentially associates with the N-(C)-A-S-H gel over the C-A-S-H gel. QUANTITATIVE F19 CONFIRMATION: Cs Lmean (Fig. 5b) = 7.8 (2%) and 7.0 (5%), both above the ANSI/ANS-16.1 floor of 6, but markedly LOWER than the framework geopolymers - fly-ash (Stanojevic 10.3/9.9; Jang ~10) and metakaolin (Nevin Cs 13.2/13.3). The calcium-silicate gel holds Cs less well than the aluminosilicate framework.

ACTION

Q9 CLOSED. Ingested 2 Pool B Cs LI rows via _from_komljenovic2020 (42 -> 44 rows / 6 sources), matrix_class slag_blend, from_figure. Note for the eventual paper: Cs leachability index now stratifies cleanly by binder chemistry - framework MK ~13 > framework FA ~10 > Ca-rich slag ~7.5 - the immobilisation-side echo of the adsorption-side framework-Al finding (F27). Komljenovic is the Provis group; same lineage as Stanojevic, Nevin, Geddes.

medium **F30 — Qian 2001 (Q10) closed - Sr/Cs distribution ratios from a scanned 2001 paper, read visually; supports the framework-Al story from the Ca side** **[resolved]**

EVIDENCE

Batch-4, Q10 acquisition. Qian, Sun & Tay, J. Nucl. Mater. 299 (2001) 199-204, doi:10.1016/S0022-3115(01)00700-0. SCANNED PDF (no text layer) - values read visually from the page-4 body text. Aluminium-rich alkali-activated slag (AAS) and M-AAS (AAS + metakaolin + zeolite + NaOH-treated attapulgite clay), a Ca-rich C-A-S-H binder. Distribution ratio Rd (0.002 M CsCl/SrCl₂, 7 d): AAS Cs 3.9e3, Sr 6.4e3; M-AAS Cs 6.4e3, Sr 1.2e4. Adding Al-rich metakaolin/clay RAISES Rd for both ions (M-AAS > AAS by 67-88%), and both beat OPC by 1.5-3x - i.e. more aluminosilicate exchange capacity = more uptake, even in a calcium-silicate binder. This is the framework-Al mechanism seen from the Ca side. UNIT FLAG: the paper labels Rd "mol/g", but Rd is a distribution coefficient whose magnitude (3.9e3-1.2e4) is consistent with mL/g. Stored in kd_mL_g with the caveat recorded, not silently "corrected".

ACTION

Q10 CLOSED. Ingested 4 Pool A KD rows (AAS/M-AAS x Cs/Sr) via _from_qian2001 (133 -> 137 rows / 13 sources), sorbent_class AAM (slag C-A-S-H, not framework). Fifth on-disk/visual recovery of the batch; second successful visual read of a scanned/figure-locked source this session.

medium

F31 — Arbel-Haddad 2022 mined - 6 Cs cumulative-leached rows for MK framework geopolymers; DOI unverifiable from the AAM and flagged as such [resolved]

EVIDENCE

Batch-4. Arbel-Haddad, Harnik, Schlosser & Goldbourt, "Cesium immobilization in metakaolin-based geopolymers elucidated by 133Cs solid state NMR spectroscopy" (J. Nucl. Mater., ~2022). Metakaolin geopolymer, Si/Al = 1 (very Al-rich framework), Na activator with Cs substituting Na at 2.5-100%. Table 4 gives the cumulative fraction of Cs leached (48 h, two-step): 2.5% 0.044, 5% 0.008, 7.5% 0.004, 25% 0.006, 50% 0.056, 100% 0.139. Leaching is LOWEST (best immobilisation) at 5-25% Cs and rises steeply at high loading - a non-monotonic loading effect. The 133Cs NMR resolves Cs SITES (structural water / framework-bound), not a Si framework descriptor, so this yields a target but no ARI. DATA-INTEGRITY NOTE: no DOI is present in this author-accepted manuscript (every DOI in the file is a bibliography reference). Provenance is recorded as a flagged citation string, NOT a guessed DOI, per the standing no-fabrication rule.

ACTION

Ingested 6 Pool B rows via _from_arbelhaddad2022 (44 -> 50 rows / 7 sources), introducing the CUMULATIVE_LEACHED_FRACTION retention type (lower = better). Provenance flagged "(DOI-unverified-AAM)"; confirm and correct the DOI when the published version is available. Loading recorded as null (Cs% is a molar Na substitution, not a wt%).

low

F32 — O'Donoghue 2026 assessed but NOT ingested - a purely structural study with no sorption or retention target on any sample [resolved]

EVIDENCE

Batch-4. O'Donoghue, Geddes, Wilkinson, Stennett, Kim, Iuga, Hayes & Walkley, "High-field multinuclear MAS NMR and synchrotron XANES reveal the influence of strontium salt chemistry on geopolymer nanostructure", Dalton Trans. 55 (2026) 10673, doi:10.1039/d6dt00775a (the Sheffield group; same lineage as Nevin/Geddes). It varies the Sr SALT (Sr(OH)2.8H2O, SrCO3, Sr(NO3)2, SrSO4) and Sr LOADING (Sr/Al 0.025-0.125) at FIXED Si/Al = 1.5, and reports full Q4(mAl) deconvolutions (SI Tables S1-S4, 3 d and 28 d) plus XANES. But there is NO quantitative Sr target anywhere - no capacity, K_d, leachability index or diffusivity (searched: all zero). Every row would be descriptor-only.

ACTION

NOT INGESTED, by the same discipline as D7 (do not pad a pool with rows that have no usable target). Unlike Nevin's structure-only rows - which sit alongside companion rows that DO carry an LI - none of O'Donoghue's samples have a target, and Si/Al is fixed so even the descriptor does not vary across the design. It remains a useful structural reference (how Sr salt chemistry and SrCO3 formation perturb the K-A-S-H framework) and is catalogued in papers/MANIFEST.md. If a Sr uptake/retention measurement on these exact samples ever appears, revisit.

medium

F33 — Foams 2020 (Petlitchkaia) extracted at last - and the bare framework foam out-capacities the ferrocyanide-functionalised one in pure water [resolved]

EVIDENCE

Petlitchkaia et al., J. Cleaner Production 269 (2020) 122400, doi:10.1016/j.jclepro.2020.122400 (CEA/Poulesquen). The batch-3 deferred item (mangled text layer), now read visually. Two metakaolin K-geopolymer foams: GF (bare) and FGF (functionalised with K2Cu(Fe(CN)6), a Prussian-blue analogue). Cs capacity is matrix-dependent, so THREE distinct quantities appear and are kept apart: isotherm MAX in pure water (GF 250, FGF 175 mg/g); single-point uptake at 100 ppm (GF 80, FGF 65 - a kinetics endpoint, NOT ingested); fresh-water capacity with competitive K/Na/Mg/Ca (GF 150, FGF 100). The isotherm reaches and passes saturation (capacity declines beyond Cs_eq 3.92e-3 mol/L), so the F14/theta screen is satisfied - these are genuine maxima, not extrapolations. MECHANISM SPLIT worth noting: GF removes Cs by FRAMEWORK ion exchange (Na<->Cs); FGF adds ferrocyanide (K<->Cs). In pure water the BARE FRAMEWORK out-capacities the functionalised foam (250 > 175 mg/g); FGF wins only on SELECTIVITY (Kd ~3.5e5 vs ~3.5e4 mL/g). Consistent with the framework-Al story - the aluminosilicate exchange sites carry the raw capacity; functionalisation trades some for selectivity.

ACTION

Ingested 4 Pool A rows via _from_petlitchkaia2020 (137 -> 141 rows / 14 sources): 2 pure-water maxima + 2 fresh-water (competing_ion flagged). Single-point 80/65 not ingested (redundant kinetics endpoint). from_figure. Batch-3 Foams deferral CLOSED. FGF is a functionalised sorbent - flag before pooling its capacity with bare framework geopolymers.

Appendix B

Pool A — every adsorption row

All 141 rows across 14 sources. ‘—’ = not reported for that row. Generated directly from the pipeline; nothing hand-entered.

#	source	sorbent	adsorbate	class	cap (mg/g)	type	Si/Al	BET	Al ^{IV} %	provenance
1	literature_co mpilation	Na-MKBFS	Cs	radionuc lide	75.8	langmuir_ qmax	2.13	—	—	El-Naggar_2018
2	literature_co mpilation	Na-MKBFS	Cs	radionuc lide	80.5	langmuir_ qmax	2.13	—	—	El-Naggar_2018
3	literature_co mpilation	Na-MKBFS	Cs	radionuc lide	82.2	langmuir_ qmax	2.13	—	—	El-Naggar_2018
4	literature_co mpilation	K-MKBFS	Cs	radionuc lide	42.2	langmuir_ qmax	2.4	—	—	El-Naggar_2018
5	literature_co mpilation	K-MKBFS	Cs	radionuc lide	48	langmuir_ qmax	2.4	—	—	El-Naggar_2018
6	literature_co mpilation	K-MKBFS	Cs	radionuc lide	61.5	langmuir_ qmax	2.4	—	—	El-Naggar_2018
7	literature_co mpilation	MU-PMK80	Cs	radionuc lide	59.6	langmuir_ qmax	1.43	—	—	Hamed_2025
8	literature_co mpilation	MU-PMK80	Eu	other	18.6	langmuir_ qmax	1.43	—	—	Hamed_2025
9	literature_co mpilation	MU-PMK80	Cs	radionuc lide	59.4	langmuir_ qmax	1.43	—	—	Hamed_2025
10	literature_co mpilation	MU-PMK80	Eu	other	20	langmuir_ qmax	1.43	—	—	Hamed_2025
11	literature_co mpilation	MU-PMK80	Cs	radionuc lide	57.7	langmuir_ qmax	1.43	—	—	Hamed_2025
12	literature_co mpilation	MU-PMK80	Eu	other	21.6	langmuir_ qmax	1.43	—	—	Hamed_2025
13	literature_co mpilation	MU-PMK80	Cs	radionuc lide	59.6	langmuir_ qmax	1.43	—	—	Hamed_2025
14	literature_co mpilation	MU-PMK80	Cs	radionuc lide	59.4	langmuir_ qmax	1.43	—	—	Hamed_2025
15	literature_co mpilation	MU-PMK80	Cs	radionuc lide	57.7	langmuir_ qmax	1.43	—	—	Hamed_2025
16	literature_co mpilation	MU-PMK80	Cs	radionuc lide	59.6	langmuir_ qmax	1.43	—	—	Hamed_2025

#	source	sorbent	adsorbate	class	cap (mg/g)	type	Si/Al	BET	Al ^{IV} %	provenance
17	literature_compilation	FA-Geopolymer	Cs	radionuclide	112	langmuir_qmax	2	—	—	Tian_2019
18	literature_compilation	FA-Geopolymer	Sr	radionuclide	24.2	langmuir_qmax	2	—	—	Tian_2019
19	literature_compilation	Zeolite-Geopolymer	Cs	radionuclide	87.9	langmuir_qmax	2	—	—	Zheng_2023
20	literature_compilation	M/SZMs	Cs	radionuclide	104	freundlich_qmax	2	—	—	Lei_2021
21	literature_compilation	M/SZMs	Sr	radionuclide	54.9	langmuir_qmax	2	—	—	Lei_2021
22	oulu2026	Si20 Al1 Na1	NH4+	analogue	2.13	single_point_q	20	249	—	doi:10.1016/j.matd
23	oulu2026	Si15 Al1 Na1	NH4+	analogue	4.27	single_point_q	15	242	—	doi:10.1016/j.matd
24	oulu2026	Si10 Al1 Na1	NH4+	analogue	5.47	single_point_q	10	206	—	doi:10.1016/j.matd
25	oulu2026	Si5Al1Na1	NH4+	analogue	7	single_point_q	5	132	—	doi:10.1016/j.matd
26	oulu2026	Si4Al1Na1	NH4+	analogue	8.07	single_point_q	4	109	—	doi:10.1016/j.matd
27	oulu2026	Si3Al1Na1	NH4+	analogue	9.13	single_point_q	3	71.6	—	doi:10.1016/j.matd
28	oulu2026	Si2Al1Na1	NH4+	analogue	11.5	single_point_q	2	31	—	doi:10.1016/j.matd
29	oulu2026	Si1Al1Na1	NH4+	analogue	13.7	single_point_q	1	8.27	—	doi:10.1016/j.matd
30	oulu2026	Si20 Al1 Na1 Ca21	NH4+	analogue	7.73	single_point_q	20	6.37	—	doi:10.1016/j.matd
31	oulu2026	Si15 Al1 Na1 Ca16	NH4+	analogue	6.87	single_point_q	15	6.04	—	doi:10.1016/j.matd
32	oulu2026	Si10 Al1 Na1 Ca11	NH4+	analogue	5.33	single_point_q	10	5.14	—	doi:10.1016/j.matd
33	oulu2026	Si5 Al1 Na1 Ca6	NH4+	analogue	4.53	single_point_q	5	3.52	—	doi:10.1016/j.matd
34	oulu2026	Si4 Al1 Na1 Ca5	NH4+	analogue	5.93	single_point_q	4	6.92	—	doi:10.1016/j.matd
35	oulu2026	Si3 Al1 Na1 Ca4	NH4+	analogue	6.67	single_point_q	3	10.3	—	doi:10.1016/j.matd
36	oulu2026	Si2 Al1 Na1 Ca3	NH4+	analogue	8.87	single_point_q	2	13.2	—	doi:10.1016/j.matd
37	oulu2026	Si1 Al1 Na1 Ca2	NH4+	analogue	11	single_point_q	1	14.1	—	doi:10.1016/j.matd
38	oulu2026	Si4 Al1 Na1 Ca1	NH4+	analogue	6.53	single_point_q	4	2.79	—	doi:10.1016/j.matd
39	oulu2026	Si4 Al1 Na1 Ca2	NH4+	analogue	6.8	single_point_q	4	1.03	—	doi:10.1016/j.matd
40	oulu2026	Si4 Al1 Na1 Ca3	NH4+	analogue	8.27	single_point_q	4	2.7	—	doi:10.1016/j.matd
41	oulu2026	Si4 Al1 Na1 Ca4	NH4+	analogue	8.67	single_point_q	4	7.95	—	doi:10.1016/j.matd
42	oulu2026	Si4 Al1 Na1 Ca5	NH4+	analogue	5.93	single_point_q	4	6.92	—	doi:10.1016/j.matd
43	oulu2026	Si20 Al1 Na1	methylene_blue	dye	90.3	single_point_q	20	249	—	doi:10.1016/j.matd
44	oulu2026	Si15 Al1 Na1	methylene_blue	dye	87.6	single_point_q	15	242	—	doi:10.1016/j.matd
45	oulu2026	Si10 Al1 Na1	methylene_blue	dye	77.2	single_point_q	10	206	—	doi:10.1016/j.matd
46	oulu2026	Si5Al1Na1	methylene_blue	dye	56.6	single_point_q	5	132	—	doi:10.1016/j.matd
47	oulu2026	Si4Al1Na1	methylene_blue	dye	48.4	single_point_q	4	109	—	doi:10.1016/j.matd
48	oulu2026	Si3Al1Na1	methylene_blue	dye	36.7	single_point_q	3	71.6	—	doi:10.1016/j.matd
49	oulu2026	Si2Al1Na1	methylene_blue	dye	14.3	single_point_q	2	31	—	doi:10.1016/j.matd
50	oulu2026	Si1Al1Na1	methylene_blue	dye	3.57	single_point_q	1	8.27	—	doi:10.1016/j.matd
51	oulu2026	Si20 Al1 Na1 Ca21	methylene_blue	dye	27	single_point_q	20	6.37	—	doi:10.1016/j.matd
52	oulu2026	Si15 Al1 Na1 Ca16	methylene_blue	dye	25.2	single_point_q	15	6.04	—	doi:10.1016/j.matd
53	oulu2026	Si10 Al1 Na1 Ca11	methylene_blue	dye	21.6	single_point_q	10	5.14	—	doi:10.1016/j.matd
54	oulu2026	Si5 Al1 Na1 Ca6	methylene_blue	dye	14.2	single_point_q	5	3.52	—	doi:10.1016/j.matd
55	oulu2026	Si4 Al1 Na1 Ca5	methylene_blue	dye	20.3	single_point_q	4	6.92	—	doi:10.1016/j.matd
56	oulu2026	Si3 Al1 Na1 Ca4	methylene_blue	dye	26.8	single_point_q	3	10.3	—	doi:10.1016/j.matd
57	oulu2026	Si2 Al1 Na1 Ca3	methylene_blue	dye	31.4	single_point_q	2	13.2	—	doi:10.1016/j.matd
58	oulu2026	Si1 Al1 Na1 Ca2	methylene_blue	dye	34.9	single_point_q	1	14.1	—	doi:10.1016/j.matd
59	oulu2026	Si4 Al1 Na1 Ca1	methylene_blue	dye	11.2	single_point_q	4	2.79	—	doi:10.1016/j.matd
60	oulu2026	Si4 Al1 Na1 Ca2	methylene_blue	dye	9.3	single_point_q	4	1.03	—	doi:10.1016/j.matd
61	oulu2026	Si4 Al1 Na1 Ca3	methylene_blue	dye	11.5	single_point_q	4	2.7	—	doi:10.1016/j.matd
62	oulu2026	Si4 Al1 Na1 Ca4	methylene_blue	dye	27.2	single_point_q	4	7.95	—	doi:10.1016/j.matd
63	oulu2026	Si4 Al1 Na1 Ca5	methylene_blue	dye	20.3	single_point_q	4	6.92	—	doi:10.1016/j.matd
64	oulu2026	Si20 Al1 Na1	rhodamine_6G	dye	53.7	single_point_q	20	249	—	doi:10.1016/j.matd
65	oulu2026	Si15 Al1 Na1	rhodamine_6G	dye	28	single_point_q	15	242	—	doi:10.1016/j.matd

#	source	sorbent	adsorbate	class	cap (mg/g)	type	Si/Al	BET	Al ^{IV} %	provenance
66	oulu2026	Si10 Al1 Na1	rhodamine_6G	dye	13.5	single_poi nt_q	10	206	—	doi:10.1016/j.matd e...
67	oulu2026	Si5Al1Na1	rhodamine_6G	dye	7.74	single_poi nt_q	5	132	—	doi:10.1016/j.matd e...
68	oulu2026	Si4Al1Na1	rhodamine_6G	dye	6.29	single_poi nt_q	4	109	—	doi:10.1016/j.matd e...
69	oulu2026	Si3Al1Na1	rhodamine_6G	dye	5.93	single_poi nt_q	3	71.6	—	doi:10.1016/j.matd e...
70	oulu2026	Si2Al1Na1	rhodamine_6G	dye	5.21	single_poi nt_q	2	31	—	doi:10.1016/j.matd e...
71	oulu2026	Si1Al1Na1	rhodamine_6G	dye	2.31	single_poi nt_q	1	8.27	—	doi:10.1016/j.matd e...
72	oulu2026	Si20 Al1 Na1 Ca21	rhodamine_6G	dye	36.3	single_poi nt_q	20	6.37	—	doi:10.1016/j.matd e...
73	oulu2026	Si15 Al1 Na1 Ca16	rhodamine_6G	dye	33.4	single_poi nt_q	15	6.04	—	doi:10.1016/j.matd e...
74	oulu2026	Si10 Al1 Na1 Ca11	rhodamine_6G	dye	27.6	single_poi nt_q	10	5.14	—	doi:10.1016/j.matd e...
75	oulu2026	Si5 Al1 Na1 Ca6	rhodamine_6G	dye	12.4	single_poi nt_q	5	3.52	—	doi:10.1016/j.matd e...
76	oulu2026	Si4 Al1 Na1 Ca5	rhodamine_6G	dye	14.6	single_poi nt_q	4	6.92	—	doi:10.1016/j.matd e...
77	oulu2026	Si3 Al1 Na1 Ca4	rhodamine_6G	dye	17.1	single_poi nt_q	3	10.3	—	doi:10.1016/j.matd e...
78	oulu2026	Si2 Al1 Na1 Ca3	rhodamine_6G	dye	18.9	single_poi nt_q	2	13.2	—	doi:10.1016/j.matd e...
79	oulu2026	Si1 Al1 Na1 Ca2	rhodamine_6G	dye	20.4	single_poi nt_q	1	14.1	—	doi:10.1016/j.matd e...
80	oulu2026	Si4 Al1 Na1 Ca1	rhodamine_6G	dye	9.18	single_poi nt_q	4	2.79	—	doi:10.1016/j.matd e...
81	oulu2026	Si4 Al1 Na1 Ca2	rhodamine_6G	dye	7.74	single_poi nt_q	4	1.03	—	doi:10.1016/j.matd e...
82	oulu2026	Si4 Al1 Na1 Ca3	rhodamine_6G	dye	10.3	single_poi nt_q	4	2.7	—	doi:10.1016/j.matd e...
83	oulu2026	Si4 Al1 Na1 Ca4	rhodamine_6G	dye	19.3	single_poi nt_q	4	7.95	—	doi:10.1016/j.matd e...
84	oulu2026	Si4 Al1 Na1 Ca5	rhodamine_6G	dye	14.6	single_poi nt_q	4	6.92	—	doi:10.1016/j.matd e...
85	katada2024	FAU 6.1	Cs	radionuclide	73.1	iec_meq_g	1.37	—	—	10.1021/acs.lang mui...
86	katada2024	LTA 6.7	Cs	radionuclide	23.9	iec_meq_g	1.12	—	—	10.1021/acs.lang mui...
87	katada2024	MFI 1.2	Cs	radionuclide	133	iec_meq_g	12.3	—	—	10.1021/acs.lang mui...
88	katada2024	YFI 1.3	Cs	radionuclide	110	iec_meq_g	11.1	—	—	10.1021/acs.lang mui...
89	katada2024	MOR 1.5	Cs	radionuclide	191	iec_meq_g	9.88	—	—	10.1021/acs.lang mui...
90	katada2024	MOR 1.7	Cs	radionuclide	211	iec_meq_g	8.37	—	—	10.1021/acs.lang mui...
914	tarnovsky202	Kaolin	Cs	radionuclide	13.3	langmuir_ qmax	1.03	9	—	10.15407/hftp15.0 1....
924	tarnovsky202	Kaolin	Sr	radionuclide	14.9	langmuir_ qmax	1.03	9	—	10.15407/hftp15.0 1....
934	tarnovsky202	GP 1	Cs	radionuclide	233	langmuir_ qmax	1.95	26	—	10.15407/hftp15.0 1....
944	tarnovsky202	GP 1	Sr	radionuclide	48.2	langmuir_ qmax	1.95	26	—	10.15407/hftp15.0 1....
954	tarnovsky202	GP 2	Cs	radionuclide	124	langmuir_ qmax	1.59	88	—	10.15407/hftp15.0 1....
964	tarnovsky202	GP 2	Sr	radionuclide	57	langmuir_ qmax	1.59	88	—	10.15407/hftp15.0 1....
97	varon2025	K-0.5-16	Sr	radionuclide	—	kd_mL_g	1.14	72	96	10.1016/j.clema.2 02...
98	varon2025	K-1-16	Sr	radionuclide	—	kd_mL_g	1.55	163	93.8	10.1016/j.clema.2 02...
99	varon2025	K-1.2-16	Sr	radionuclide	—	kd_mL_g	1.67	188	94	10.1016/j.clema.2 02...
1000	varon2025	K-1.5-16	Sr	radionuclide	—	kd_mL_g	1.79	212	91.4	10.1016/j.clema.2 02...
1001	varon2025	K-1.2-8 (inferred)	Sr	radionuclide	—	kd_mL_g	1.83	15	88	10.1016/j.clema.2 02...
1002	varon2025	K-1.2-12 (inferred)	Sr	radionuclide	—	kd_mL_g	1.63	182	92.4	10.1016/j.clema.2 02...
1003	varon2025	Na-1.2-16 (inferred)	Sr	radionuclide	—	kd_mL_g	1.67	38	84.7	10.1016/j.clema.2 02...
1004	lin2026	ACSr	Sr	radionuclide	222	iec_meq_g	—	—	—	10.3390/ceramics 902...
1005	lin2026	NCGO	Cs	radionuclide	54.5	iec_meq_g	—	39.2	—	10.3390/ceramics 902...
1006	lin2026	NiGO	Cs	radionuclide	46.5	iec_meq_g	—	38.8	—	10.3390/ceramics 902...
1007	lin2026	CoGO	Cs	radionuclide	45.2	iec_meq_g	—	29.3	—	10.3390/ceramics 902...
1008	lin2026	Ba-4A	Sr	radionuclide	167	iec_meq_g	—	—	—	10.3390/ceramics 902...
1009	lin2026	CuFC/4A	Cs	radionuclide	86.4	iec_meq_g	—	5.25	—	10.3390/ceramics 902...
110	tian2019_esp or	Zeolite A (from fly ash)	Cs	radionuclide	282	langmuir_ qmax	0.821	121	—	10.1007/s11356- 019-...

#	source	sorbent	adsorbate	class	cap (mg/g)	type	Si/Al	BET	Al ^{IV} %	provenance
111	tian2019_espr	Zeolite A (from fly ash)	Sr	radionuclide	169	langmuir_qmax	0.821	121	—	10.1007/s11356-019-...
112	xiang2021	GF300	Cs	radionuclide	162	single_point_q	1.02	53.8	—	10.1016/j.jece.2021...
113	xiang2021	GF400	Cs	radionuclide	164	single_point_q	1.02	49.4	—	10.1016/j.jece.2021...
114	xiang2021	GF500	Cs	radionuclide	186	single_point_q	1.02	45.8	—	10.1016/j.jece.2021...
115	xiang2021	GF600	Cs	radionuclide	192	single_point_q	1.02	37.8	—	10.1016/j.jece.2021...
116	xiang2021	GF600	Cs	radionuclide	216	langmuir_qmax	1.02	37.8	—	10.1016/j.jece.2021...
117	zhang2021	Flyash-geomaterial	Cs	radionuclide	89.3	langmuir_qmax	—	—	—	10.3390/app11188407
118	zhang2021	Slag-geomaterial	Cs	radionuclide	44.5	langmuir_qmax	—	—	—	10.3390/app11188407
119	baek2018	chabazite	Cs	radionuclide	1.25e+03	langmuir_qmax	1.5	—	—	10.1016/j.micromeso...
120	baek2018	stilbite	Cs	radionuclide	76.3	langmuir_qmax	3.88	—	—	10.1016/j.micromeso...
121	baek2018	heulandite	Cs	radionuclide	59.5	langmuir_qmax	3.48	—	—	10.1016/j.micromeso...
122	kurumisawa2021	K11	Cs	radionuclide	41.9	langmuir_qmax	—	—	—	10.1617/s11527-021-...
123	kurumisawa2021	K9	Cs	radionuclide	56.5	langmuir_qmax	—	—	—	10.1617/s11527-021-...
124	kurumisawa2021	0.66K	Cs	radionuclide	70.4	langmuir_qmax	—	—	—	10.1617/s11527-021-...
125	kurumisawa2021	N11	Cs	radionuclide	96.6	langmuir_qmax	—	—	—	10.1617/s11527-021-...
126	kurumisawa2021	KN11	Cs	radionuclide	52.2	langmuir_qmax	—	—	—	10.1617/s11527-021-...
127	varon_leached2026	K-0.5-16 (leached 96h)	Sr	radionuclide	51.7	experimental_max	1.27	109	82.4	10.1016/j.oceram.20...
128	varon_leached2026	K-1-16 (leached 96h)	Sr	radionuclide	35	experimental_max	1.52	181	92.1	10.1016/j.oceram.20...
129	varon_leached2026	K-1.2-16 (leached 96h)	Sr	radionuclide	35.9	experimental_max	1.63	199	91.9	10.1016/j.oceram.20...
130	varon_leached2026	K-1.5-16 (leached 96h)	Sr	radionuclide	21	experimental_max	1.77	221	93.6	10.1016/j.oceram.20...
131	varon_leached2026	K-1.2-8 (leached 96h)	Sr	radionuclide	44.7	experimental_max	1.63	28	88.8	10.1016/j.oceram.20...
132	varon_leached2026	K-1.2-12 (leached 96h)	Sr	radionuclide	48.2	experimental_max	1.63	220	90.4	10.1016/j.oceram.20...
133	varon_leached2026	Na-1.2-16 (leached 96h)	Sr	radionuclide	24.5	experimental_max	1.63	44	85.8	10.1016/j.oceram.20...
134	qian2001	AAS	Cs	radionuclide	—	kd_mL_g	—	—	—	10.1016/S0022-3115(...
135	qian2001	AAS	Sr	radionuclide	—	kd_mL_g	—	—	—	10.1016/S0022-3115(...
136	qian2001	M-AAS	Cs	radionuclide	—	kd_mL_g	—	—	—	10.1016/S0022-3115(...
137	qian2001	M-AAS	Sr	radionuclide	—	kd_mL_g	—	—	—	10.1016/S0022-3115(...
138	petlitckaia2020	GF (geopolymer foam, pure water)	Cs	radionuclide	250	experimental_max	—	—	—	10.1016/j.jclepro.2...
139	petlitckaia2020	FGF (functionalised geopolymer foam, pure water)	Cs	radionuclide	175	experimental_max	—	—	—	10.1016/j.jclepro.2...
140	petlitckaia2020	GF (geopolymer foam, fresh water)	Cs	radionuclide	150	experimental_max	—	—	—	10.1016/j.jclepro.2...
141	petlitckaia2020	FGF (functionalised geopolymer foam, fresh water)	Cs	radionuclide	100	experimental_max	—	—	—	10.1016/j.jclepro.2...

Appendix C

Pool B — every immobilisation row

All 50 rows across 7 sources. ‘target’ distinguishes a leachability index (higher = better) from an ingress diffusivity; the two are never pooled. ‘cens.’ marks a left-censored (below-detection, best-performing) result.

#	source	matrix	class	nuclide	target	value	cens.	ARI	poros.%	provenance
1	kim2026	NaSr1	na_geopolymer	Sr	leachability index	15.5		—	—	10.1016/j.cemconres...
2	kim2026	NaSr2	na_geopolymer	Sr	leachability index	14.8		—	—	10.1016/j.cemconres...
3	kim2026	NaSr3	na_geopolymer	Sr	leachability index	16		—	—	10.1016/j.cemconres...
4	kim2026	KSr1	k_geopolymer	Sr	leachability index	—	left	—	—	10.1016/j.cemconres...

#	source	matrix	class	nuclide	target	value	cens.	ARI	poros.%	provenance
5	kim2026	KSr2	k_geopolymer	Sr	leachability index	—	left	—	—	10.1016/j.cemconres...
6	kim2026	KSr3	k_geopolymer	Sr	leachability index	16		—	—	10.1016/j.cemconres...
7	nevin2026	C_0	k_geopolymer	none	—	—		2.81	—	10.1039/d5ta90211h
8	nevin2026	Sr_1	k_geopolymer	Sr	leachability index	18.7		2.75	—	10.1039/d5ta90211h
9	nevin2026	Sr_3	k_geopolymer	Sr	leachability index	19.1		2.8	—	10.1039/d5ta90211h
10	nevin2026	Cs_1	k_geopolymer	Cs	leachability index	13.3		2.74	—	10.1039/d5ta90211h
11	nevin2026	Cs_3	k_geopolymer	Cs	leachability index	13.2		2.85	—	10.1039/d5ta90211h
12	nevin2026	C_0	k_geopolymer	none	—	—		2.81	—	10.1039/d5ta90211h
13	nevin2026	Sr_1	k_geopolymer	Sr	—	—		2.93	—	10.1039/d5ta90211h
14	nevin2026	Sr_3	k_geopolymer	Sr	—	—		2.81	—	10.1039/d5ta90211h
15	nevin2026	Cs_1	k_geopolymer	Cs	—	—		2.8	—	10.1039/d5ta90211h
16	nevin2026	Cs_3	k_geopolymer	Cs	—	—		2.76	—	10.1039/d5ta90211h
17	kurumisawa2021	K11	k_geopolymer	Cs	d ingress cm2 s	4.70e-08		—	—	10.1617/s11527-021-...
18	kurumisawa2021	K11	k_geopolymer	Cs	d ingress cm2 s	4.00e-08		—	—	10.1617/s11527-021-...
19	kurumisawa2021	K9	k_geopolymer	Cs	d ingress cm2 s	7.00e-09		—	—	10.1617/s11527-021-...
20	kurumisawa2021	K9	k_geopolymer	Cs	d ingress cm2 s	4.80e-09		—	—	10.1617/s11527-021-...
21	kurumisawa2021	0.66K	k_geopolymer	Cs	d ingress cm2 s	4.40e-08		—	—	10.1617/s11527-021-...
22	kurumisawa2021	0.66K	k_geopolymer	Cs	d ingress cm2 s	4.00e-08		—	—	10.1617/s11527-021-...
23	kurumisawa2021	N11	na_geopolymer	Cs	d ingress cm2 s	8.30e-09		—	—	10.1617/s11527-021-...
24	kurumisawa2021	N11	na_geopolymer	Cs	d ingress cm2 s	4.70e-09		—	—	10.1617/s11527-021-...
25	kurumisawa2021	KN11	k_geopolymer	Cs	d ingress cm2 s	1.90e-08		—	—	10.1617/s11527-021-...
26	kurumisawa2021	KN11	k_geopolymer	Cs	d ingress cm2 s	2.00e-08		—	—	10.1617/s11527-021-...
27	jang2016	FA1 geopolymer	fly_ash_geopolymer	Cs	leachability index	10		—	29.9	10.1016/j.jhazmat.2...
28	jang2016	FA1 geopolymer	fly_ash_geopolymer	Sr	leachability index	14.1		—	29.9	10.1016/j.jhazmat.2...
29	jang2016	FA2 geopolymer	fly_ash_geopolymer	Cs	leachability index	10.1		—	33.9	10.1016/j.jhazmat.2...
30	jang2016	FA2 geopolymer	fly_ash_geopolymer	Sr	leachability index	14.1		—	33.9	10.1016/j.jhazmat.2...
31	jang2016	FA3 geopolymer	fly_ash_geopolymer	Cs	leachability index	10.2		—	31.9	10.1016/j.jhazmat.2...
32	jang2016	FA3 geopolymer	fly_ash_geopolymer	Sr	leachability index	14.4		—	31.9	10.1016/j.jhazmat.2...
33	jang2016	FA3+BFS 90:10	slag_blend	Cs	leachability index	7.1		—	39.7	10.1016/j.jhazmat.2...
34	jang2016	FA3+BFS 90:10	slag_blend	Sr	leachability index	11.5		—	39.7	10.1016/j.jhazmat.2...
35	jang2016	FA3+BFS 70:30	slag_blend	Cs	leachability index	8		—	31.9	10.1016/j.jhazmat.2...
36	jang2016	FA3+BFS 70:30	slag_blend	Sr	leachability index	12.7		—	31.9	10.1016/j.jhazmat.2...
37	jang2016	FA3+BFS 50:50	slag_blend	Cs	leachability index	8		—	25.9	10.1016/j.jhazmat.2...
38	jang2016	FA3+BFS 50:50	slag_blend	Sr	leachability index	12.1		—	25.9	10.1016/j.jhazmat.2...
39	jang2016	Portland cement	portland_cement	Cs	leachability index	7		—	28.3	10.1016/j.jhazmat.2...
40	jang2016	Portland cement	portland_cement	Sr	leachability index	9.8		—	28.3	10.1016/j.jhazmat.2...
41	stanojevic2025	G-FA-2%Cs	fly_ash_geopolymer	Cs	leachability index	10.3		—	—	10.1016/j.conbuildm...
42	stanojevic2025	G-FA-5%Cs	fly_ash_geopolymer	Cs	leachability index	9.9		—	—	10.1016/j.conbuildm...
43	komljenovic2020	AABFS-2%Cs	slag_blend	Cs	leachability index	7.8		—	—	10.1016/j.jhazmat.2...
44	komljenovic2020	AABFS-5%Cs	slag_blend	Cs	leachability index	7		—	—	10.1016/j.jhazmat.2...
45	arbelhaddad2022	MK-GP 2.5%Cs	na_geopolymer	Cs	clf	0.044		—	—	Arbel-Haddad 2022 J...
46	arbelhaddad2022	MK-GP 5%Cs	na_geopolymer	Cs	clf	0.008		—	—	Arbel-Haddad 2022 J...
47	arbelhaddad2022	MK-GP 7.5%Cs	na_geopolymer	Cs	clf	0.004		—	—	Arbel-Haddad 2022 J...
48	arbelhaddad2022	MK-GP 25%Cs	na_geopolymer	Cs	clf	0.006		—	—	Arbel-Haddad 2022 J...
49	arbelhaddad2022	MK-GP 50%Cs	na_geopolymer	Cs	clf	0.056		—	—	Arbel-Haddad 2022 J...
50	arbelhaddad2022	MK-GP 100%Cs	na_geopolymer	Cs	clf	0.139		—	—	Arbel-Haddad 2022 J...

Appendix D

Decisions register

The 16 standing project decisions, each with its reason. These are the rules the pipeline enforces.

#	Decision	Reason
D 1	Drop Si_Al and Density from physical data	F1 - transplanted, not measured
D 2	Re-scope M3b to cross-system transfer	Different material system; no viscosity; 92 mixes
D 3	Repo private, all-rights-reserved	Unpublished research
D 4	Copyrighted PDFs excluded from git	Copyright + 80 MB; manifest tracks them
D 5	Do not use Feasibility_Ranges	F6 - anti-correlated with ground truth
D 6	Kd kept in its own column, never in capacity_mg_g	Different physical quantity
D 7	Reject Tian supplement's 24-row table	No composition; would pad without features
D 8	Quarantine Na-MK, K-MK and all Niu 2022 rows	F7 - not in stated sources
D 9	No subagents for extraction or analysis	Missing project context lets silent errors in
D 10	Pre-register NMR acceptance at r>=0.9	Deconvolution is under-determined; stops retroactive tuning
D 11	ARI claims are gated on Q4-rich framework sorbents	F19 - Ca-bearing gels have no framework; descriptor is meaningless there, not merely weak
D 12	Never pool sorbent structural classes in a correlation	F19 - pooling hid ARI +0.932 and -0.108 behind a single +0.536
D 13	Run adsorption capacity and leach resistance as TWO parallel pools, never merged	PI decision 2026-07-20 on F20. Removal-from-solution and immobilisation are different applications with different targets; the descriptor is available only in the second. Consistent with D12.
D 15	Re-read every triage-rejected paper inline before trusting the rejection	F24 - Kurumisawa was rejected on keyword balance, but its Cs capacity lives only in a figure, which keyword counts cannot see
D 16	Never compute ARI from a deconvolution whose m-assignment is ambiguous	F23 - Vettese's Table 4 admits ARI 2.289 or 0.861, a 2.7x fork
D 14	Index the corpus by CONTENT, not filename	F13/F20 - Kim 2026 and Nevin 2026, the two most descriptor-rich papers held, were invisible to a filename-based check because their filenames are not in Latin script

Appendix E

Glossary of chemical terms

Term	Meaning
Alkali-activated material (AAM)	Binder made by activating an aluminosilicate precursor with a strong alkali; geopolymers are the aluminium-rich, low-calcium end of this family.
Al ^{IV} / framework aluminium	Aluminium in tetrahedral coordination within the network; each site carries a negative charge balanced by an exchangeable cation. The predictive descriptor of this project.
ARI (Al-richness index)	Mean number of Al neighbours per Si, $\Sigma_m m \cdot I_m / 100$, from ²⁹ Si NMR speciation; an NMR-based measure of framework aluminium.
BET surface area	Specific surface area from N ₂ physisorption (m ² g ⁻¹). Predicts dye (physisorption) uptake but <i>not</i> cation (ion-exchange) uptake.
Distribution coefficient K _d	Ratio of sorbed to solution concentration (mL g ⁻¹); a measure of affinity, distinct from a capacity.
Effective diffusivity D _e	Rate constant for a nuclide leaching out of a waste form (cm ² s ⁻¹); lower is better.
Leachability index (LI, L)	−log ₁₀ (D _e); ANSI/ANS 16.1 ranking of immobilisation. Higher = better; ≥6 is the acceptance floor.
Langmuir Q _{max}	Fitted monolayer saturation capacity (mg g ⁻¹); meaningful only if the experiment approached saturation (the θ screen).
Q ⁿ (mAl)	A silicon environment with <i>n</i> bridging oxygens, <i>m</i> of them to aluminium; from ²⁹ Si MAS NMR.
R _L / θ	Langmuir separation factor; θ = 1 − R _L is the fraction of saturation reached, used to screen fitted capacities.
Si/Al ratio	Molar silicon-to-aluminium ratio; lower means more charge-balancing sites.
Network modifier (Ca)	Calcium disrupts the aluminosilicate framework toward low-connectivity C–A–S–H, where framework descriptors lose meaning.